

AMMonitor Audio Tools - Cheat Sheet (Part 1)

1 127.0.0.1:3313

AMMonitor

2

Audio Tagger

Recording Filters & Settings 3

3 +

Recording-level Tags 5

recording_non_taxa_bbox

☐ highlight

☐ corrupted

Taxon Tags 6

Select taxon

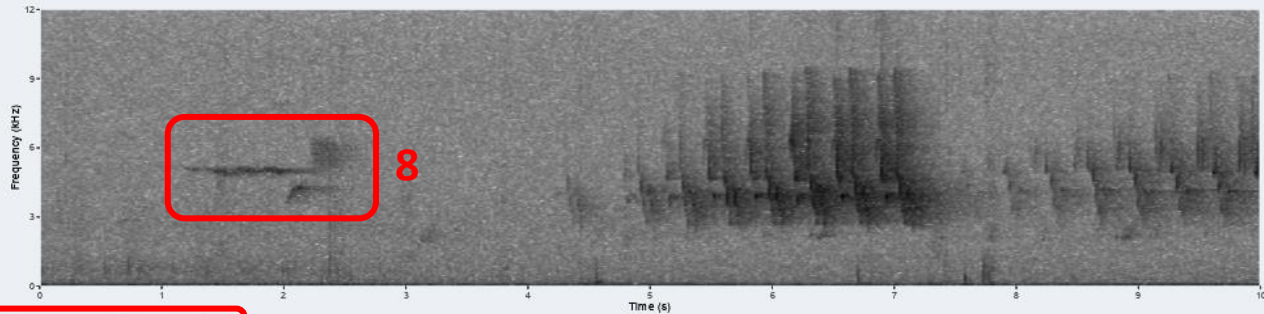
btnw

vocalization_type

☐ template

Recording 1 of 1: FileName: 1.wav; Location: locationA; Date/Time: 2023-05-05 06:02:49 4

0:00 / 0:23 7



Previous Spec Next Spec 9

Previous file Next file Save Labels 10

Taxon Tags 11

ID	person	taxon	Start time (s)	End time (s)	Min freq (KHz)	Max freq (KHz)	libraryid	item	value_num
No rows found									

File Tags 12

person	fk_medialistid	item	value_num
No rows found			

Delete selected annotation(s) 13

Verify Tags 14

Agree

Disagree

Clear

AMMonitor Audio Tools - Cheat Sheet (Part 1)

1. Do not use your web browser's "back" button. The browser "refresh" button restarts the app.
2. (tagger only) From left to right, these buttons will
 - Copy the tags from the previous image. Shortcut: SHIFT+C
 - Add a new tag (from selected options below). Shortcut: SHIFT-A
3. Press the "+" plus sign to see audio filter and settings options (see Cheat Sheet - Part 2)
4. Recording metadata is displayed in text above each image, including the file name, location, and date/time.
5. Recording-level tags pertain to the whole recording (and not a particular taxon).
6. Taxon tags are annotations and annotags that pertain to a particular taxon. After selecting a taxon, all relevant annotags are displayed below. The "no-species" taxon is used to indicate the absence of a target species.
7. The HTML audio player can be used to play recordings, pause them, set the playback time, adjust the volume, and download the file.
8. To create a bounding box, click and drag on the spectrogram to define a box. Unsaved bounding boxes have a yellow border, and saved boxes have a red border. To remove an un-saved bounding box, double-click inside the box. A single-click on the spectrogram will set the playback time, as specified.
9. These buttons can be used to move to the next/previous spectrogram, when the window size is shorter than the recording length.
10. These buttons, or the arrow keys → and ←, change recordings. The "Save Labels" button (not visible in "viewer" mode) is clickable only when there are unsaved labels (annotations/annotags/mediatags/verifications).
11. Taxon Tags include all annotations and annotags associated with the recording. Tags are filtered such that: in viewer mode, all tags are visible; in tagger mode, only tags made by the currently signed-in user are displayed; in verifier mode, only tags made by other users are displayed. When in viewer mode, model outputs can be included here by selecting this option from the recording settings. When performing model verifications, only modeloutputs are displayed. A *verified* column is displayed when using the verifiers.
12. File Tags include all mediatags associated with the recording. This table behaves similarly to the Taxon Tags table with respect to what is visible when using each of the different audio tools (i.e., viewer, tagger, verifiers).
13. Delete annotations/annotags/mediatags selected from the above tables.
14. These buttons can be used to add verifications that agree or disagree with either annotations/annotags/mediatags (Annotation Verifications), or with modeloutputs (Model Verifications). Verifications can be deleted with the "Clear" button. If now rows are selected to specify which tags to verify, the verification is applied to all tags. Keyboard shortcuts are the "plus" (+) key to "Agree" with a label or the "minus" key "-" to "Disagree" with a label.



AMMonitor Audio Tools - Cheat Sheet (Part 2)

1

-

Recording Filters & Settings

Audio Filters

4

Select a Location:

all

Select Date Range (Defaults to all):

2023-05-05 to 2023-05-05

Select Taxa:

all

Exclude annotated by

☒ NA
☐ Me
☐ Anyone

Exclude if verified by

☒ NA
☐ Me
☐ Anyone

Model Filters

5

Select model

all

Model Value:

☐ Less than Value

Visit Filters

6

Select a location to show visits.

visit_date	total_recordings	total_annotated
No rows found		

Jump to Recording

7

Select Recording:

Apply Filters

8

1 recordings found

Audio/Spectrogram Settings

2

Audio Directory Path/URL:

C:\Users\jclarke\AppData\Local\Temp\Rtmp2zvQNW\demo_869863233\recordings\

Cache size (# of recordings)

1000000

Spectrogram Length (s):

10

Window Name

hanning

Overlap

0

Spectrogram Frequency Range (kHz):

0 to 12

☐ Show Waveform
☒ Interpolate
☐ Show Model Outputs

Auto-save Rate (# of recordings)

1

Window Length

512

Spectrogram Color Palette:

reverse.gray.colors.2

Zero Padding

0

Audio Modifications

3

☐ Remove noise

Audio Frequency Range (kHz):

0 to 12

Amplitude Filter (%)

0 to 10

Reset Audio Filters

AMMonitor Audio Tools - Cheat Sheet (Part 2)

1. Use the minus-sign “-” to hide the available/selected recording filters and settings
2. Audio/Spectrogram settings include:
 - Defining a “root” file path or URL for reading audio files. If the filepath field for the recording is missing, this will be appended to the beginning of the filename when trying to render the recording.
 - Cache size refers to the number of image files for which metadata is stored in memory at once.
 - The auto-save rate (tagger and verifier modes only) controls how often the database is modified when changes are made. Set the auto-save rate to “1” to save automatically after each change.
 - Spectrogram settings, colormaps, and whether to include the waveform.
 - (viewer mode only) Whether to show modeloutputs along with annotations/annotags/mediatags.
3. Audio modifications allow you to see/hear changes from amplitude, frequency, and noise reduction filters.
4. Recordings can be filtered by location, date range, and taxon. For the taxa filter, only labels made by the currently logged-in user will be displayed. There are also options for excluding recordings tagged (tagger mode only) or verified (verifier mode only) by the currently logged-in user (i.e., “Me”), or by anyone.
5. Model filters allow filtering of model outputs by model name and/or model value, returning modeloutputs that are greater than or less than a specified value.
6. After selecting a location, the visits from that location will appear here. Selecting a visit will filter just recordings associated with the selected visit. The total number of recordings from each visit from the selected location are shown, along with the total number of recordings that the currently logged-in user has annotated.
7. Go to a specific recording by file name (max. of 5,000 displayed).
8. When un-applied filters have been selected, a warning message will be displayed. Press the “Apply Filters” after changing filter settings to run a new query based on the selected options. The number of images returned by the selected filters are shown here.

Note: AMMonitor allows setting defaults to be saved by using configuration files stored in the “settings” directory of an AMMonitor project. For more information on how to add default settings, and for video tutorials, run the AMMonitor tutorial:

```
learnr::run_tutorial("annotations", package = "AMMonitor")
```



Remote monitoring of biodiversity
in an adaptive framework with R

AMMonitor 2.0.0 • Updated: 2024-06-18
Learn more at <https://code.usgs.gov/vtcfwru/ammonitor>